NNWA Nutrition and Wellness Academy
पोषण और कल्याण अकादमी

EDITION 1 · 2026

# Protein, *decoded.*

How much you need, where to find it in everyday Indian food, and how to spread it across your day.

## A plain-language guide

From NNWA (Nutrition & Wellness Academy). General information for healthy adults, not medical advice.

01

INSIDE THIS GUIDE

# Seven short *chapters*.

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02 How much you need 05

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HOW TO USE THIS GUIDE

Read it once, then use the workbook in chapter 07 for a week. Every figure is listed with its source on page 23.

BEFORE YOU START

General information for healthy adults, not medical advice. If you are pregnant, breastfeeding, under 18, older, unwell or training heavily, speak to a qualified healthcare professional.

01

# What protein *does.*

Protein is in every cell of your body. Before the numbers, here is what it actually does and why food has to supply part of it every day.

01 What protein does
02 How much you need
03 Where to find it
04 Spread it across your day
05 Myths, checked
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01 · WHAT PROTEIN DOES

# Six jobs, *every* day.

Protein is made of building blocks called amino acids. The body uses about 20 of them. It can make some itself; the rest, called essential amino acids, must come from food.

**9**

essential amino acids adults must get from food

01

### Builds and repairs

Muscle, skin, hair, nails and organs are rebuilt from protein all the time.

02

### Runs reactions

Enzymes, which drive digestion and metabolism, are proteins.

03

### Sends signals

Some hormones, such as insulin, are proteins.

04

### Defends

Antibodies that help fight infection are proteins.

05

### Carries

Haemoglobin, the protein in red blood cells, carries oxygen.

06

### Gives energy

About 4 kcal per gram, though the body prefers carbohydrate and fat for fuel.

WHY DAILY MATTERS

Unlike fat, the body keeps no dedicated store of protein to draw on. Body proteins are broken down and rebuilt every day, so a steady supply from meals keeps that cycle running.

02

# How much do you *need?*

One number covers most healthy adults. This chapter gives you that number, a table to find yours, and the situations where it changes.

01 What protein does

02 How much you need

03 Where to find it

04 Spread it across your day

05 Myths, checked

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02 · HOW MUCH YOU NEED

# One number to *remember*.

**0.83** g per kg of body weight, a day

This is the Recommended Dietary Allowance (RDA) for healthy Indian adults set by ICMR-NIN. An RDA covers nearly all healthy people in a group, so many people need slightly less.

| QUICK MATHS | | | | |
|---|---|---|---|---|
| | **Weight** in kg | **× 0.83** the RDA | **= grams** per day | **60 → 50** kg to g, example |

## When the number changes

| | |
|---|---|
| **Pregnancy and breastfeeding** | Needs go up. ICMR-NIN sets separate, higher amounts for each stage. |
| **Children and teenagers** | Needs are set by age and growth stage, not by the adult figure. |
| **Older adults** | Protein matters for keeping muscle with age; the right amount is best set individually. |
| **Kidney or liver disease** | Some conditions need less protein, not more. Follow your doctor's advice. |
| **Heavy training** | Athletes may need more, within limits. See chapter 05. |

Source: ICMR-NIN, Nutrient Requirements for Indians, 2020. General information only, not medical advice.

02 · FIND YOUR NUMBER

# Your weight, your *grams*.

| | | |
|---|---|---|
| 37 g | **at 45 kg** | a day |
| 42 g | **at 50 kg** | a day |
| 46 g | **at 55 kg** | a day |
| 50 g | **at 60 kg** | a day |
| 58 g | **at 70 kg** | a day |
| 66 g | **at 80 kg** | a day |

Daily protein at 0.83 g per kg, rounded to the nearest gram. Source: ICMR-NIN, Nutrient Requirements for Indians, 2020.

# 03

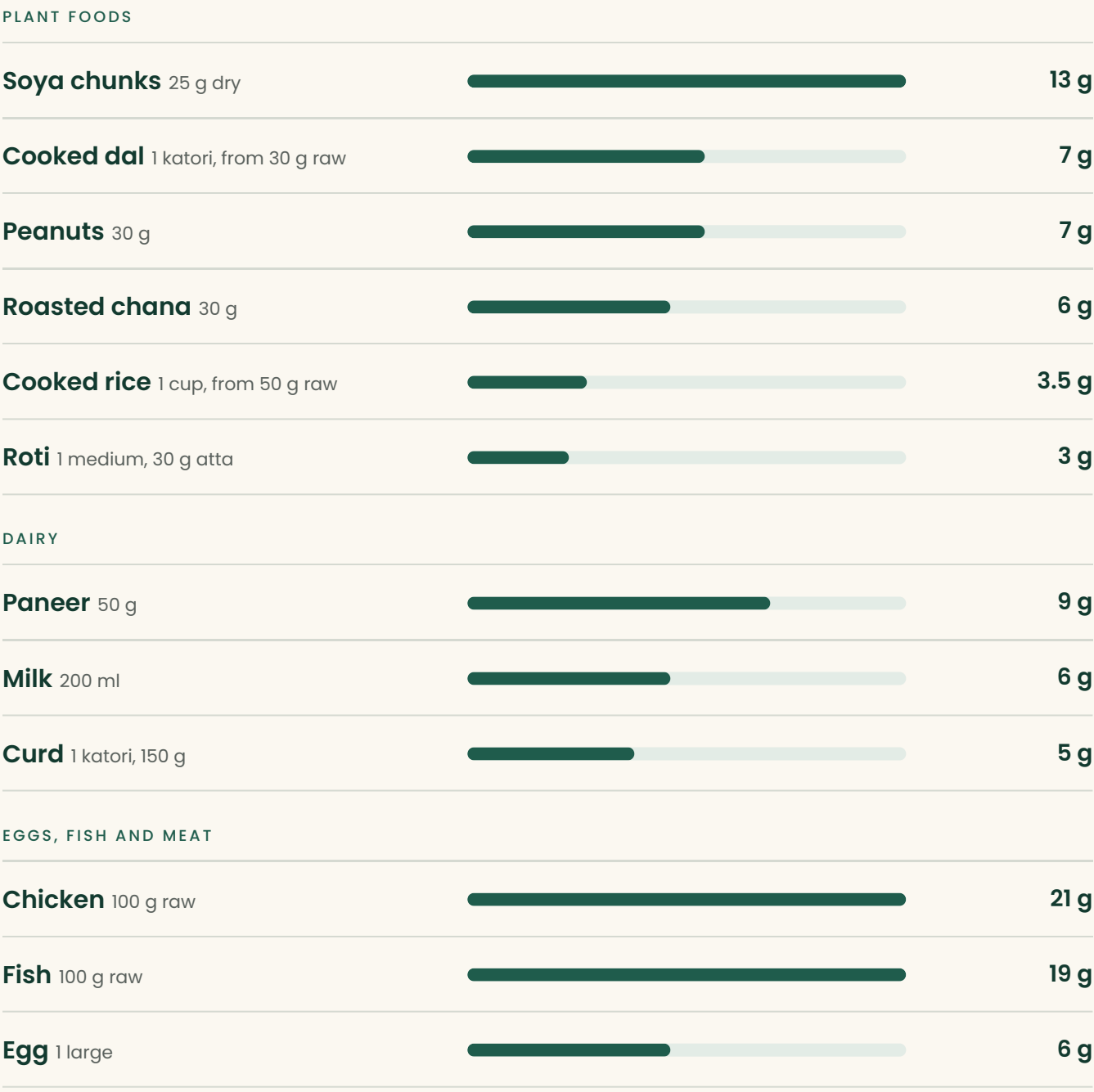
# Where to *find* it.

Protein is already in Indian kitchens: in dal, curd, paneer, eggs, soya, nuts, fish and chicken. Here is roughly how much each serving brings, and how to pair foods well.

01 What protein does
02 How much you need
03 **Where to find it**
04 Spread it across your day
05 Myths, checked
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03 · PROTEIN PER SERVING

# Roughly how much, per *serving*.




Approximate, rounded values calculated from per-100 g figures in ICMR-NIN's Indian Food Composition Tables (IFCT 2017). Actual amounts vary with variety, recipe and portion. Bars scale to 13 g; chicken and fish run past the scale.

03 · PAIR IT UP

# Better *together*.

Cereals such as rice and wheat are low in the essential amino acid lysine. Pulses are low in methionine. Eaten in the same meal, each fills the other's gap, which is why so many Indian staples already pair them.

01

### Dal and chawal

Rice with any dal.

02

### Rajma or chole with rice

Kidney beans or chickpeas.

03

### Idli or dosa with sambar

Rice and urad batter, toor dal sambar.

04

### Khichdi

Rice and moong dal in one pot.

05

### Roti with chana or dal

Wheat with pulses.

06

### Poha with peanuts

Flattened rice with a legume.

COMPLETE ON THEIR OWN

Milk, curd, paneer, eggs, fish, meat and soya provide all nine essential amino acids by themselves. Adding a little of any of them to a cereal meal also lifts its protein quality.

03 · FOOD CARD 1 OF 3

# Dal and *chawal*



| PROTEIN INFORMATION | ≈ 10 g | 1 katori | 1 cup | Paired |
|---|---|---|---|---|
| | protein | cooked dal | cooked rice | pulse + cereal |

## On the plate

- Dal, from 30 g raw: ≈ 7 g
- Rice, from 50 g raw: ≈ 3.5 g
- Add a katori of curd: ≈ 5 g

## Good to know

Rice is low in lysine and dal is low in methionine. Eaten together, each fills the other's gap, so the meal's protein is better used than either food alone.

*Try:* rajma-chawal, chole with rice, or khichdi.

Approximate values from IFCT 2017 per-100 g figures; actual amounts vary by dal, recipe and portion. Photo via Pexels.

03 · FOOD CARD 2 OF 3

# Paneer, *portioned*



| PROTEIN INFORMATION | ≈ 9 g | 50 g | Dairy | Complete |
|---|---|---|---|---|
| | protein | paneer | food group | all 9 essential |

## On the plate

- Paneer, 50 g: ≈ 9 g
- With 2 rotis: + ≈ 6 g
- Shown here in a gravy

## Good to know

Paneer is a complete protein and an easy vegetarian option. It is also rich in fat, so a measured 50 g portion keeps a meal balanced. Grilled, bhurji or in palak works better than heavy, creamy gravies every day.

*Try:* paneer bhurji with roti, paneer tikka, or palak paneer.

Approximate values from IFCT 2017 per-100 g figures; actual amounts vary by milk and recipe. Photo via Pexels.

03 · FOOD CARD 3 OF 3

# Eggs, two *ways*



| PROTEIN INFORMATION | ≈ 6 g | 1 egg | Eggs | Complete |
|---|---|---|---|---|
| | per large egg | about 50 g | food group | all 9 essential |

## On the plate

- 1 large egg: ≈ 6 g
- 2 eggs at breakfast: ≈ 12 g
- With 1 roti: + ≈ 3 g

## Good to know

An egg is one of the simplest complete proteins. The white holds a little over half the protein; the yolk carries most of the vitamins and fat, so eating the whole egg gives you both.

*Try:* a boiled egg, egg bhurji with vegetables, or an omelette with roti.

Approximate values from IFCT 2017 per-100 g figures. Photo via Pexels.

# 04

# Spread it *across* the day.

A protein food at every meal is easier than one big serving, and it fits how most Indian households already eat.

01 What protein does
02 How much you need
03 Where to find it
04 **Spread it across your day**
05 Myths, checked
06 Supplements, carefully
07 Your workbook

04 · AN EXAMPLE DAY

# Little and *often*.

One illustrative day for a 60 kg adult, whose RDA works out to about 50 g. It is an example to adapt, not a meal plan.

| | | |
|---|---|---|
| BREAKFAST | **A glass of milk with an egg or a handful of roasted chana**<br>Swap in curd, besan chilla or sprouts. | 12 g |
| LUNCH | **Two rotis, a katori of dal, a katori of curd, sabzi**<br>Cereal and pulse in the same meal. | 18 g |
| SNACK | **A small handful of peanuts or sprouts chaat**<br>Easy to carry; watch the portion. | 7 g |
| DINNER | **Rice with rajma, or 50 g paneer with sabzi**<br>Fish or chicken works here too. | 13 g |

DAY TOTAL, APPROXIMATELY

50 g



Values are approximate, from the serving table on page 09.

05

# Myths, *checked.*

Five things people often repeat about protein, set against what Indian dietary guidance and basic nutrition science say.

01 What protein does
02 How much you need
03 Where to find it
04 Spread it across your day
05 **Myths, checked**
06 Supplements, carefully
07 Your workbook

05 · MYTHS, CHECKED

# What the evidence *says*.

01

~~MORE PROTEIN ALWAYS MEANS MORE MUSCLE~~

**Training builds muscle. Protein supports it, up to a point.**

Muscle grows with resistance exercise plus enough protein and energy. ICMR-NIN's 2024 guidelines note that intakes above about 1.6 g/kg a day add no further training gains.

02

~~VEGETARIANS CANNOT GET ENOUGH PROTEIN~~

**A varied vegetarian diet can meet the RDA.**

Pulses with cereals, plus dairy, soya, nuts and seeds, cover the needs of most healthy adults.

03

~~PLANT PROTEIN DOES NOT COUNT~~

**It counts. It works best in combination.**

Single plant foods may be low in one amino acid. Pairing cereals and pulses fills the gap (page 10).

04

~~YOU NEED A SHAKE AFTER EVERY WORKOUT~~

**For most people, regular meals do the job.**

Indian dietary guidance advises against protein supplements for building muscle (chapter 06).

05

~~PROTEIN IS ONLY FOR GYM-GOERS~~

**Everyone needs it, every day.**

Repair, enzymes, immunity and oxygen transport all depend on protein, whatever your activity level.

CHECK A CLAIM YOURSELF

Ask who is saying it, whether a named source such as ICMR-NIN backs it, and whether it promises a quick result. Sound nutrition advice rarely does.

06

# Supplements, *carefully.*

Protein powders are everywhere. Here is what India's dietary guidelines say about them, and what to check if you are still considering one.

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07 Your workbook

06 · FOOD FIRST

# Start with *food*.

WHAT THE GUIDELINES SAY

ICMR-NIN's Dietary Guidelines for Indians (2024) advise against protein supplements for building muscle and recommend meeting needs through food combinations. They warn that large amounts of protein powder taken over long periods may be linked to bone mineral loss and kidney damage.

## If you are still considering one

01

### Speak to a professional first

Especially if you have any medical condition, are pregnant, or are under 18.

02

### Check the FSSAI licence number

Buy from a traceable seller, not an unlabelled or imported-without-label pack.

03

### Read the label, not the claims

Protein per serving, serving size, added sugar and the full ingredients list.

04

### Be wary of big promises

Quick muscle, weight loss or other health results on a label are a warning sign.

Source: ICMR-NIN, Dietary Guidelines for Indians, 2024. General information only, not medical advice.

07

# Your *workbook.*

Two pages to fill in. Track a week of protein habits, then keep the checklist where you will see it.

01 What protein does
02 How much you need
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07 · WEEKLY TRACKER

# A week of *habits*.

Tick a box each time the habit happens. The ticks in the first column show one example day.

| HABIT | M | T | W | T | F | S | S |
|---|---|---|---|---|---|---|---|
| **Protein at breakfast**<br>milk, curd, egg, chana | ✓ | | | | | | |
| **Protein at lunch**<br>dal, rajma, paneer, fish | ✓ | | | | | | |
| **Protein at dinner**<br>one serving, not the day's worth | ✓ | | | | | | |
| **Cereal + pulse together**<br>dal-chawal, khichdi, idli-sambar | ✓ | | | | | | |
| **Dairy or soya serving**<br>curd, milk, paneer, tofu | | | | | | | |
| **Nuts or seeds as a snack**<br>a small handful | ✓ | | | | | | |
| **Label check before supplements**<br>FSSAI number, serving, sugar | ✓ | | | | | | |

EASIEST MEAL THIS WEEK

ONE CHANGE FOR NEXT WEEK

07 · CHECKLIST

# Six habits to *keep*.

- [ ] **I know my number** — Worked out below, and checked with a professional if page 06 applies to me. — ONCE
- [ ] **Protein at every meal** — Breakfast, lunch and dinner each include a protein food. — DAILY
- [ ] **A cereal and pulse together** — Dal-chawal, khichdi, idli-sambar or roti-chana. — DAILY
- [ ] **A dairy or soya serving** — Curd, milk, paneer, tofu or soya chunks. — DAILY
- [ ] **Nuts or seeds as a snack** — A small handful, not the whole packet. — MOST DAYS
- [ ] **Label check before any supplement** — FSSAI number, serving size, added sugar, professional advice. — EVERY TIME

**My weight** _____ **kg × 0.83 =** _____ *grams a day*

SOURCES AND NOTES

# Where the numbers *come* from.


01

**ICMR-National Institute of Nutrition. Nutrient Requirements for Indians. 2020.**

Adult protein RDA of 0.83 g/kg body weight per day; separate requirements for pregnancy, lactation, children and adolescents. Pages 06, 07, 15.

02

**ICMR-National Institute of Nutrition. Dietary Guidelines for Indians. 2024.**

Protein quality through food combinations; advice against protein supplements for muscle building; the note that intakes above about 1.6 g/kg add no further training gains. Pages 10, 17, 19.

03

**ICMR-National Institute of Nutrition. Indian Food Composition Tables (IFCT). 2017.**

Per-100 g values used to calculate the approximate serving figures on pages 09 and 11 to 13 and 15. Serving figures are NNWA's rounded calculations.

04

**General physiology**

Amino acids, the nine essential amino acids, protein's roles and its energy value of about 4 kcal per gram are standard nutrition science (page 04).

05

**Photographs**

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IMPORTANT

This guide is general educational information for healthy adults. It is not medical advice and does not diagnose, treat or prevent any condition. If you are pregnant or breastfeeding, under 18, an older adult, training heavily, or living with kidney, liver or any other medical condition, consult a qualified healthcare professional before changing your diet. NNWA (Nutrition & Wellness Academy) is an education provider.

# Understand the science on your *plate.*

NNWA (Nutrition & Wellness Academy) is a 100% online academy for nutrition, wellness and fitness certification. This guide is one of a series of short, sourced nutrition basics.

EXPLORE PROGRAMMES

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